1. Network Overview

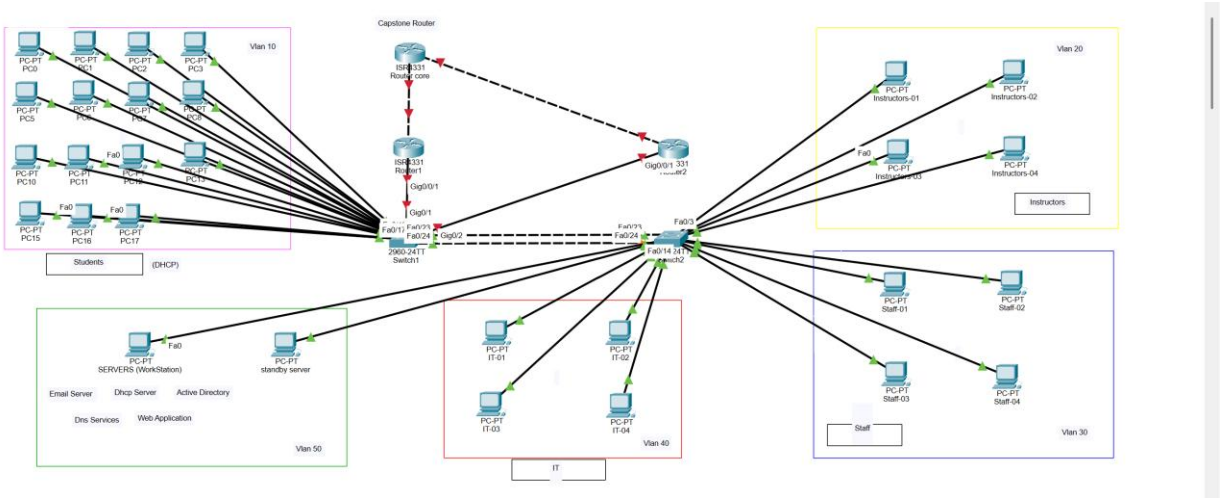
The Capstone Scenario A, on which our proposal is based, is a high school that needs a working network to support: Internet and learning access for students ,Sharing of resources and availability of instructional resources for instructors ,Administrative access and staff communication. Network service administration and IT infrastructure (AD, DNS, DHCP, internal email, etc.)

Objectives

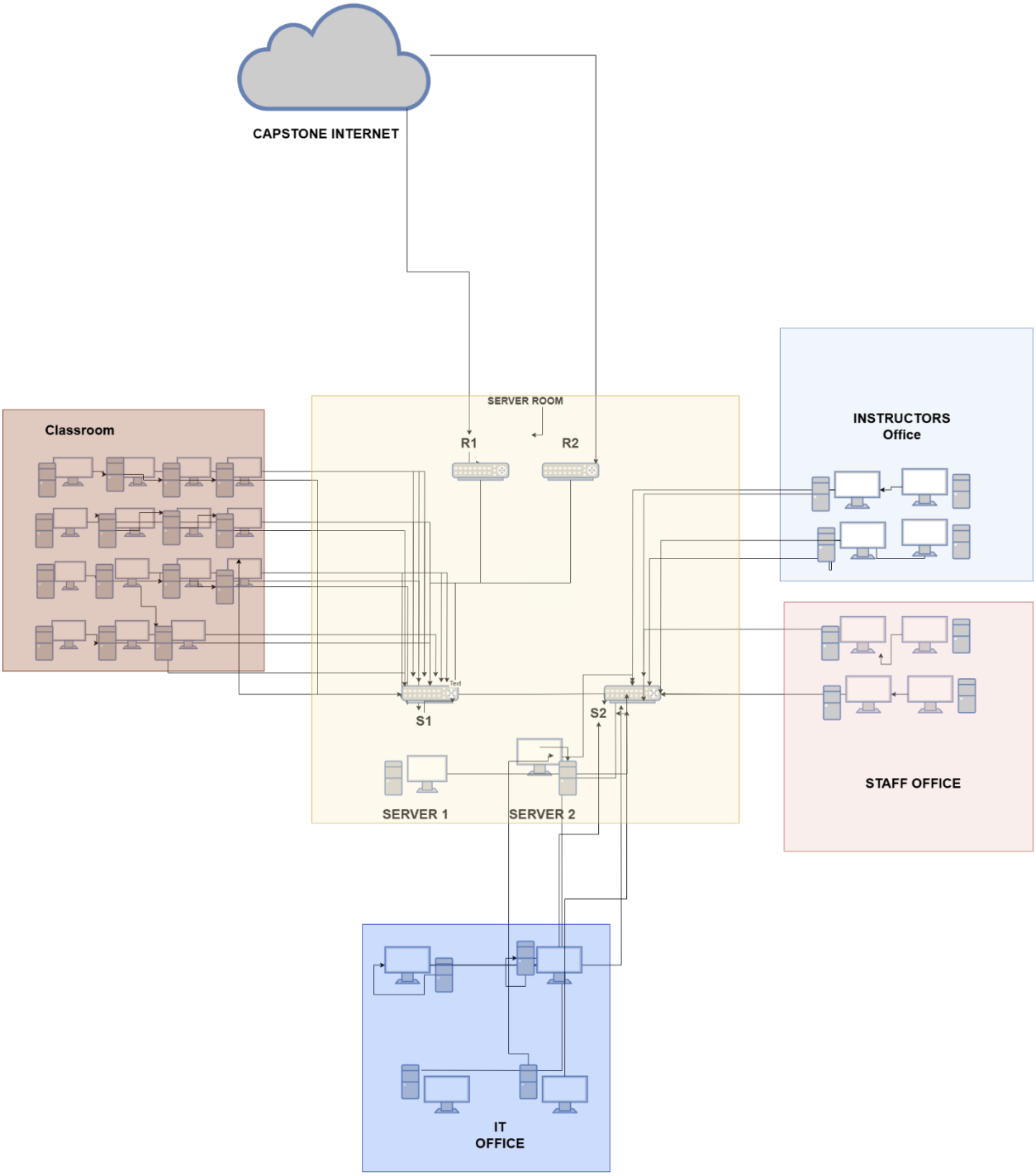
- Provide Reliable and Segmented Connectivity for All Departments
- Ensure High Availability and Redundancy
- Secure the network through Hardening and Access Control

2. Topologies

2.1 Logical Topology



2.2 Physical Topology



3. IP Addressing Schema Information

3.1 Etherchannel and Trunk Implementation

Network Devices	**Trunk Port**	**Etherchannel**
Router1	G0/1.10–G0/1.50 (ROAS)	G0/1
Router2	G0/1.10–G0/1.50 (ROAS)	G0/1
Switch1 (sw1)	Port-Channel 1	Fa0/23, Fa0/24
	Gi0/1, Gi0/2	(not bundled, trunk only)
Switch2 (Sw2)	Port-channel 1	Fa0/23, Fa0/24

3.2 ROUTER 1

IP Addressing Schema Information

Network Device	Model	Interfaces	IP address	Subnetmask
R1	Cisco 2901	G0/0(WAN)	10.128.250.122	255.255.255.0
		G0/1 (LAN Native)	192.168.1.1	255.255.255.0
		G0/1.10 (VLAN 10 - Students)	192.168.10.1	255.255.255.0
		G0/1.20 (VLAN 20 - Instructors	192.168.20.1	255.255.255.0
		G0/1.30 (VLAN 30 - Staff)	192.168.30.1	255.255.255.0
		G0/1.40 (VLAN 40 – IT)	192.168.40.1	255.255.255.0
		G0/1.50 (VLAN 50 - Servers	192.168.50.1	255.255.255.0

NOTE:

- Subinterfaces are created for Router-on-a-Stick inter-VLAN routing.
- NAT is configured with ip nat inside on G0/1 subinterfaces and ip nat outside on G0/0.
- ip helper-address is used on VLANs (e.g., 192.168.50.11) to forward DHCP requests.
- HSRP is configured on each subinterface with virtual IP .254.

Link to Router 1 Configuration router 1.txt

3.3 ROUTER 2

IP Addressing Schema Information

Network Device	Model	Interfaces	Ip Address	Subnet Mask
R2	**Cisco 2901**	**G0/0 (WAN)**	**10.128.250.123**	**255.255.255.0**
		G0/1 (LAN Native)	192.168.1.2	255.255.255.0
		G0/1.10 (VLAN 10 - Students)	192.168.10.2	255.255.255.0
		G0/1.20 (VLAN 20 - Instructors)	192.168.20.2	255.255.255.0
		G0/1.30 (VLAN 30 - Staff	192.168.30.2	255.255.255.0
		G0/1.40 (VLAN 40 - IT/Admins)	192.168.40.2	255.255.255.0
		G0/1.50 (VLAN 50 - Servers)	192.168.50.2	255.255.255.0

NOTE :

- This router is the HSRP standby with priority 110 and .2 IPs.
- It participates in OSPF area 0 with all VLANs.
- Gig0/0 is not used yet (could be used for external redundancy).
- NAT, OSPF, and HSRP are configured similarly to R1 for high availability

Link to Router2 configuration router 2.txt

3.3 Switch 1

Network Device	Model	Interfaces	IP address	Subnet Mask
S1	2960- 24TT	Fa0/1-Fa0/15 - Access (VLAN 10 (Students)	DHCP	255.255.255.0
		Fa0/23, Fa0/24 – Trunk, EtherChannel (Port-channel1)		
		Gi0/1, Gi0/2 – Trunk		

		Port-channel1 – Trunk with VLANs 10,20,30,40,50		

NOTE :

- VLAN 10: All Fa0/1 to Fa0/15 are access ports for Students.
- Trunk Links: Fa0/23 and Fa0/24 are bundled using EtherChannel (channel-group 1) and allow VLANs 10–50.
- VLAN Interface: interface Vlan10 is configured with an IP helper address pointing to the DHCP server (192.168.50.11)

Link to Switch 1 Configuration

Switch 2

Network Device	Model	Interfaces	IP Address	Subnet Mask
SW2	2960	Fa0/1 Vlan 20 (Instructors)	192.168.20.11	255.255.255.0
		Fa0/2 - Fa0/4 (Instructors)		
		Fa0/5 Vlan 30 (Staff)	192.168.30.11	255.255.255.0
		Fa0/6 - Fa0/8 Vlan 30(Staff)		
		Fa0/9	192.168.40.11	255.255.255.0
		Fa0/10 - Fa0/12 Vlan 40 (IT)		
		Fa0/13	192.168.50.11	255.255.255.0
		Fa0/14	192.168.50.12	255.255.255.0
		Fa0/23, Fa0/24 – Trunk, EtherChannel (Port-channel1)		
		Port-channel1 – Trunk (VLANs 10,20,30,40,50 allowed)		

NOTE :

- Port-channel1 bundles Fa0/23 and Fa0/24 into an EtherChannel trunk between SW1 and SW2, carrying VLANs 10–50.
- All Fa0/1–Fa0/14 are configured as access ports for the respective VLANs.

Link to Switch 2 Configuration switch 2.txt

3.4 ROUTING TABLE (R1 , R2)

Route Type	Destination	Next Hop / Interface	Note
C	192.168.10.0/24	VLAN 10 subinterface	Connected
C	192.168.20.0/24	VLAN 20 subinterface	Connected
C	192.168.30.0/24	VLAN 30 subinterface	Connected
C	192.168.40.0/24	VLAN 40 subinterface	Connected
C	192.168.50.0/24	VLAN 50 subinterface	Connected
C	192.168.1.0/24	G0/1	Connected
C	10.128.250.0/24	G0/0	Connected
S	0.0.0.0/0	G0/0	Static default route (Internet)
O	192.168.x.0/24	learned from neighbor	OSPF routes from the other router

C = Connected, S = Static, O = OSPF

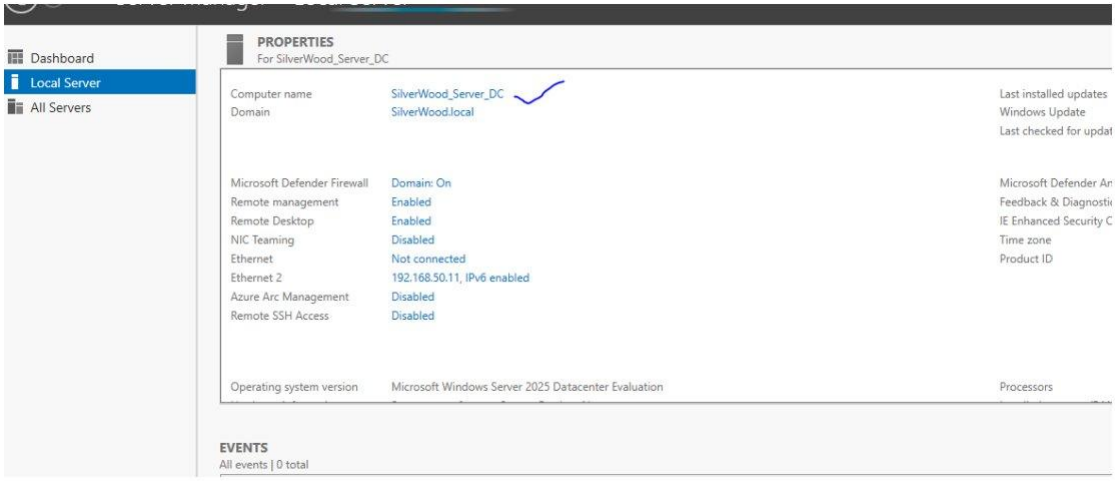
3.5 FHRP (HSRP) Table

VLAN	Subnet	HSRP Virtual IP	Active Router	Standby Router	Priority (R1/R2)
10	192.168.10.0/24	192.168.10.254	R1	R2	150 / 110
20	192.168.20.0/24	192.168.20.254	R1	R2	150 / 110
30	192.168.30.0/24	192.168.30.254	R1	R2	150 / 110
40	192.168.40.0/24	192.168.40.254	R1	R2	150 / 110
50	192.168.50.0/24	192.168.50.254	R1	R2	150 / 110

4 SERVER SERVICES

4.1 Domain Controller

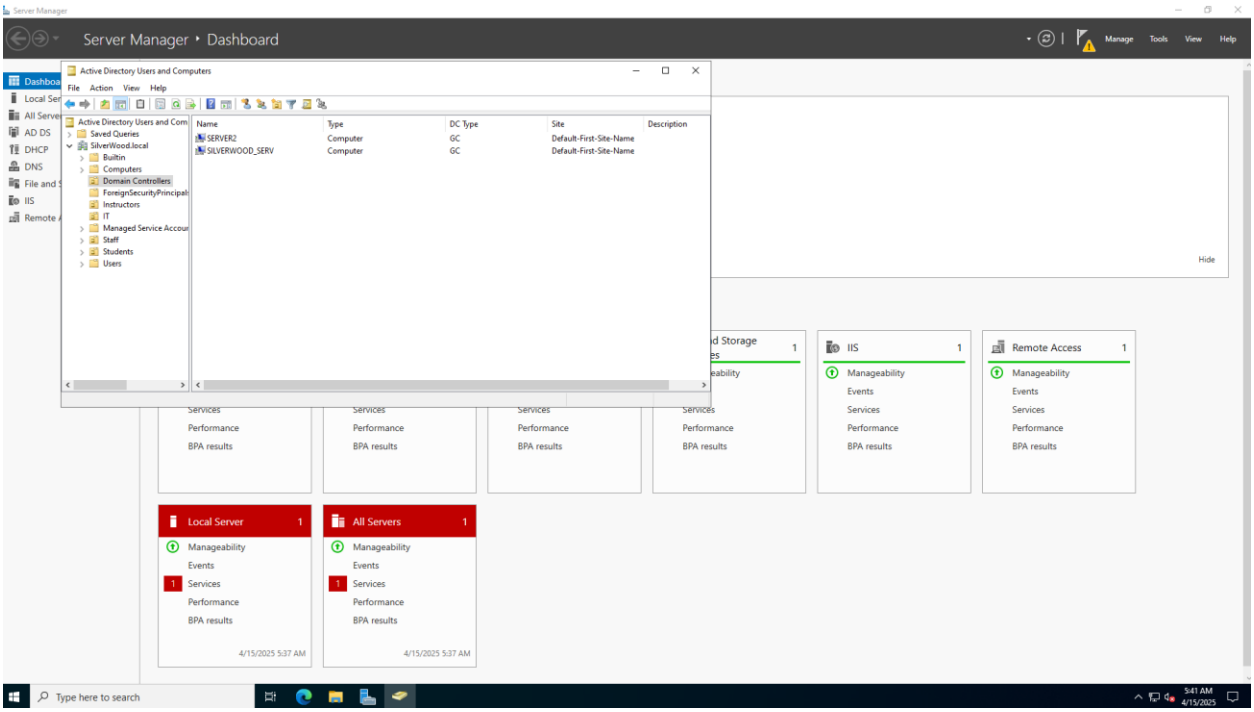
Centralizing and simplifying user identification, management, and access control throughout the school network calls for a domain controller. Using Active Directory Domain Services (AD DS), SilverWood Admins would effectively handle all student, staff, and teacher accounts from one location, therefore enforcing uniform security rules and permissions. This guarantees that particular network resources—such as departmental files, printers, or systems—only authorized users may access.



4.3 WEB SERVER AND DATABASE

This is the Link to our Webserver.
https://silverwoodschool.github.io/school-website/

4.4 SERVER AND SERVICE REDUNDANCY



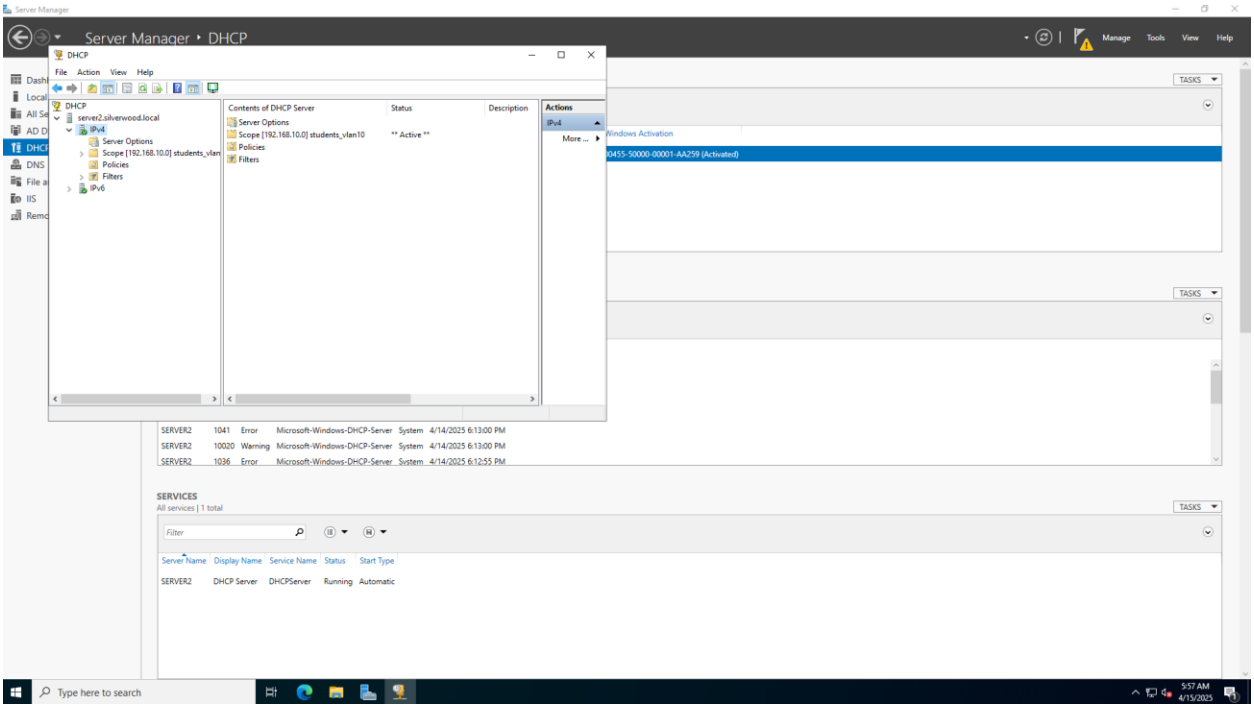
This image displays our Secondary server replicating the AD of the Primary Server

Login Details

Username – Admin

Password - AdminPass123

IP address – 192.168.50.12



This Image displays DHCP Service redundancy being displayed in our secondary server

Administrator: Command Prompt

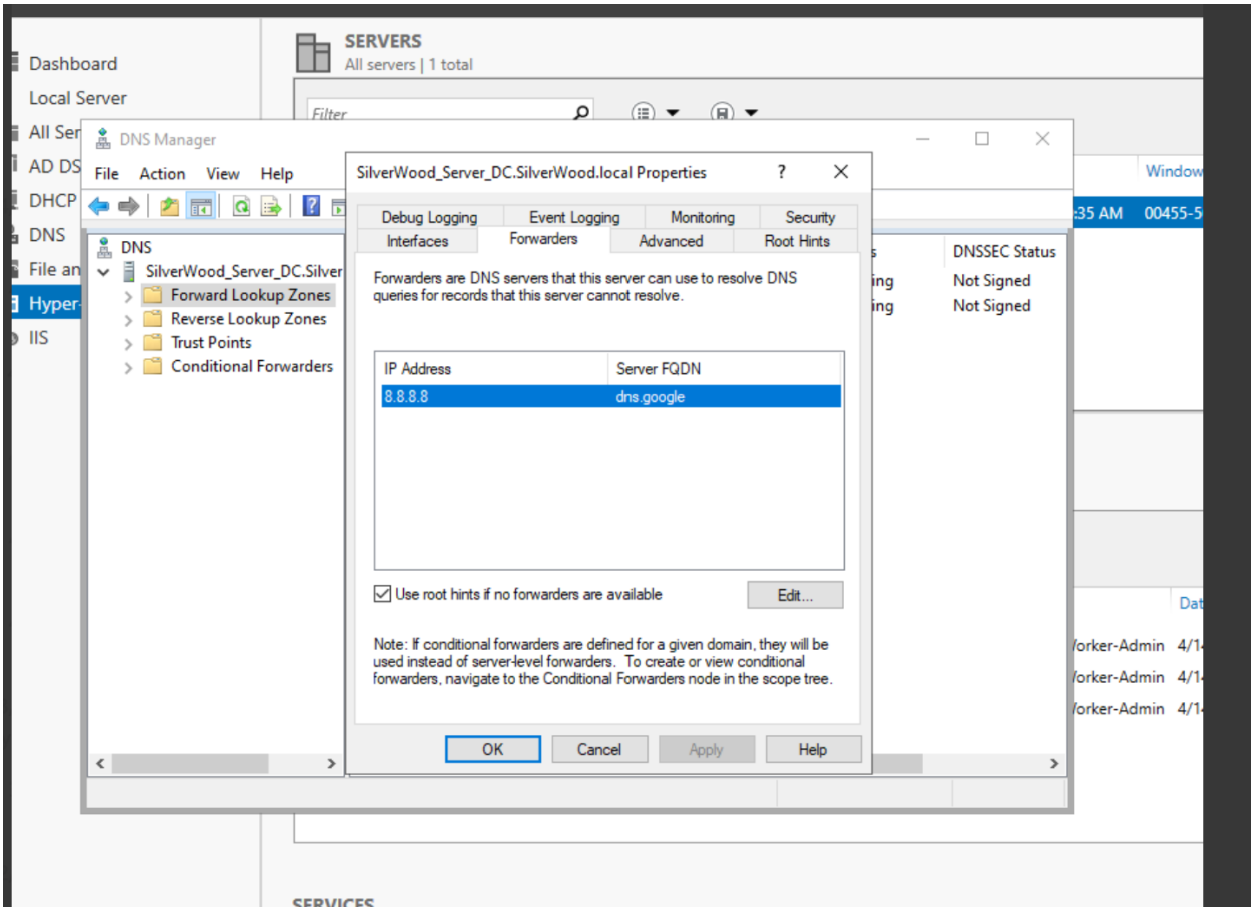
```
C:\Users\Administrator.SILVERWOOD>repadmin /showrepl

Repadmin: running command /showrepl against full DC localhost
Default-First-Site-Name\SERVER2
DSA Options: IS_GC
Site Options: (none)
DSA object GUID: e74fa1e4-d058-4a99-8980-f690ddccc26c
DSA invocationID: 95b9b194-7de6-4bbf-8b8a-6dc364ecba16

==== INBOUND NEIGHBORS ======================================

DC=SilverWood,DC=local
    Default-First-Site-Name\SILVERWOOD_SERV via RPC
        DSA object GUID: 4468d5dc-4ac3-4414-83dc-a7125c9456c2
        Last attempt @ 2025-04-15 06:05:40 was successful.

CN=Configuration,DC=SilverWood,DC=local
    Default-First-Site-Name\SILVERWOOD_SERV via RPC
        DSA object GUID: 4468d5dc-4ac3-4414-83dc-a7125c9456c2
        Last attempt @ 2025-04-15 05:50:35 was successful.

CN=Schema,CN=Configuration,DC=SilverWood,DC=local
    Default-First-Site-Name\SILVERWOOD_SERV via RPC
        DSA object GUID: 4468d5dc-4ac3-4414-83dc-a7125c9456c2
        Last attempt @ 2025-04-15 05:47:42 was successful.

DC=DomainDnsZones,DC=SilverWood,DC=local
    Default-First-Site-Name\SILVERWOOD_SERV via RPC
        DSA object GUID: 4468d5dc-4ac3-4414-83dc-a7125c9456c2
        Last attempt @ 2025-04-15 05:47:42 was successful.

DC=ForestDnsZones,DC=SilverWood,DC=local
    Default-First-Site-Name\SILVERWOOD_SERV via RPC
        DSA object GUID: 4468d5dc-4ac3-4414-83dc-a7125c9456c2
        Last attempt @ 2025-04-15 05:47:42 was successful.


C:\Users\Administrator.SILVERWOOD>
```

This Image displays our Secondary DC offering a DNS service redundancy .

4.5 Local and Global DNS

Using DNS across our capstone network project offers a number of important advantages. First, it makes access to internal services simpler by letting users link to resources under easily remembered names like Silverwood.local rather than IP addresses like 192.168.50.11.

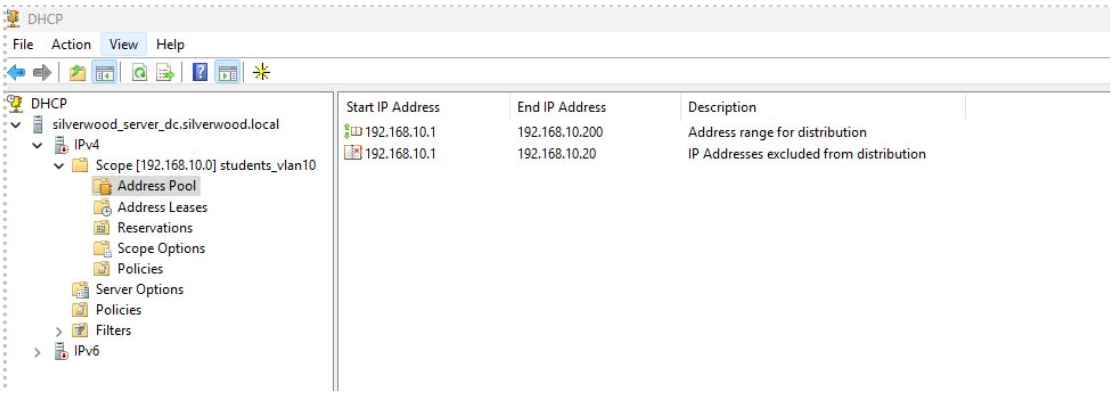
```
C:\Users\Administrator>ping www.Silverwood.local

Pinging www.Silverwood.local [192.168.50.11] with 32 bytes of data:
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.50.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping dns.silverwood.local

Pinging dns.silverwood.local [192.168.50.11] with 32 bytes of data:
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128
Reply from 192.168.50.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.50.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>
```



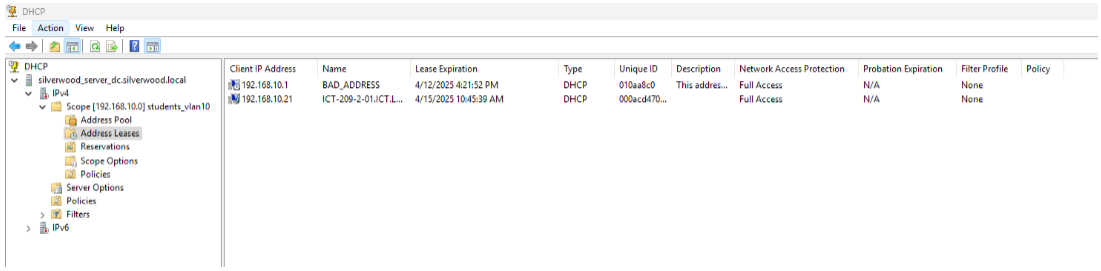
4.6 DHCP

DHCP Configuration for VLAN 10 (Students)
The **DHCP server** (located in VLAN 50, IP: 192.168.50.11) is responsible for assigning IP addresses to clients in VLAN 10. The DHCP scope is configured as follows:

DHCP Scope Configuration: Configured a DHCP scope for 192.168.10.0 with an address pool from 192.168.10.1 to 192.168.10.200, excluding 192.168.10.1 to 192.168.10.20.



I. **DHCP Lease Monitoring**: Viewed the current IP leases assigned by DHCP, including one labeled as BAD_ADDRESS and another assigned to a specific client.



5.0 Microsoft 365 Email (Send/Receive Email)

Microsoft 365 Admin Center – User Account Creation and Role Assignment

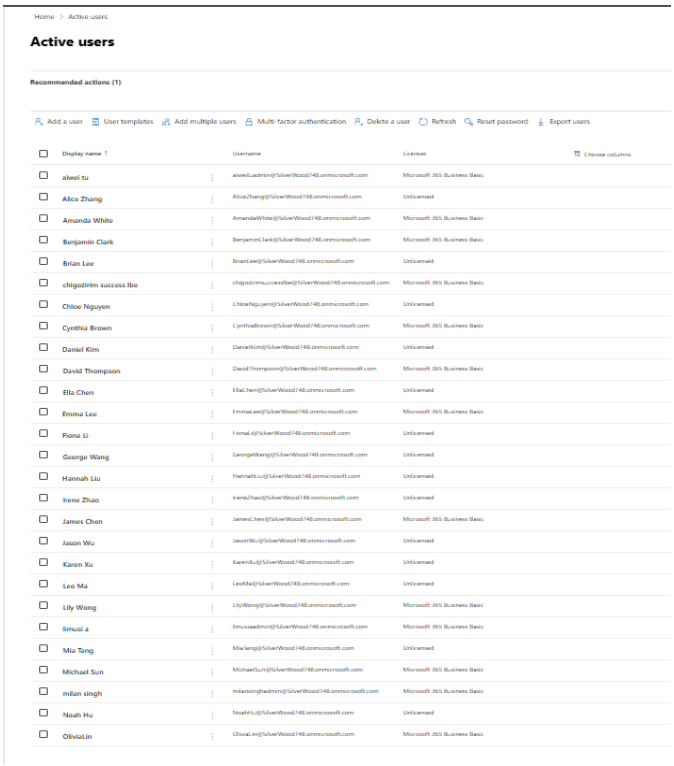
These images illustrate the process of creating new user accounts and assigning roles within the Microsoft 365 Admin Center for an organization named **SilverWood**.

1. Admin Center Overview

The first image shows the **Microsoft 365 Admin Center home screen** where an administrator can manage users, licenses, teams, and products. It provides a simplified view for quick access to common setup tasks like adding users and assigning roles.

Active Users List

- This screenshot shows the **Active Users** page, listing all currently registered users in the tenant. You can see each user's display name, username and license type

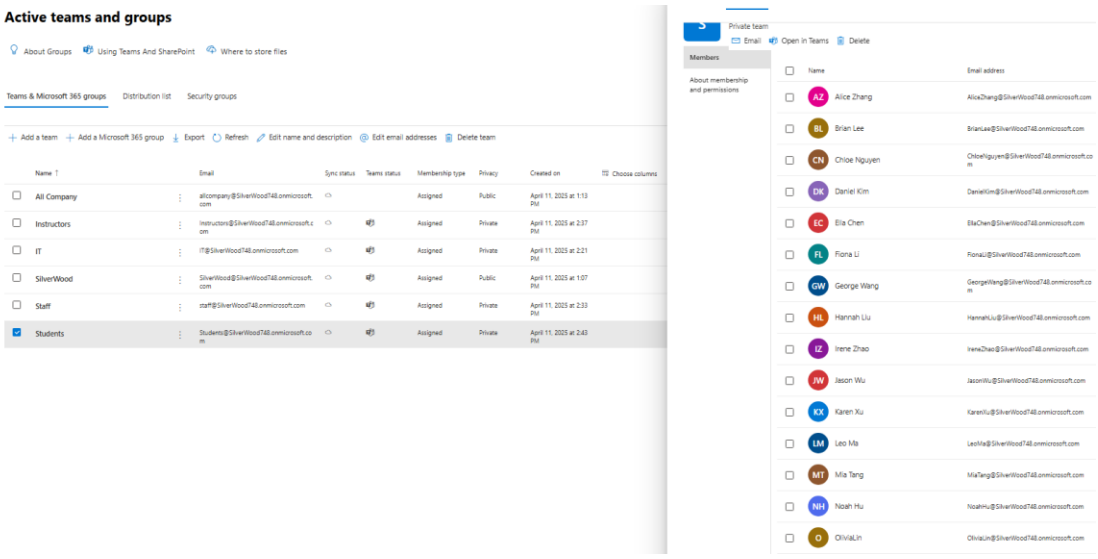


6. Microsoft 365 Admin Center – Microsoft Teams Group Creation and Setup

These screenshots demonstrate how different department-based Microsoft Teams groups are created and configured within the **SilverWood** organization using the Microsoft 365 Admin Center. This setup supports clear **organizational structure**, ensures **controlled access**, and allows **efficient internal communication** within each department using Microsoft Teams.

Group Management Overview

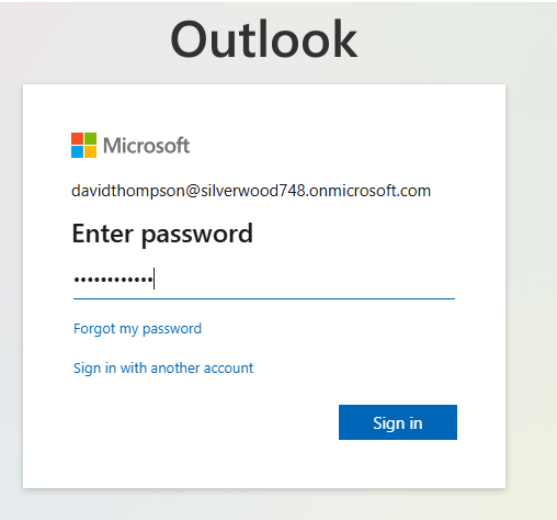
- This image displays the **"Active Teams and Groups"** interface within Microsoft 365 Admin Center.



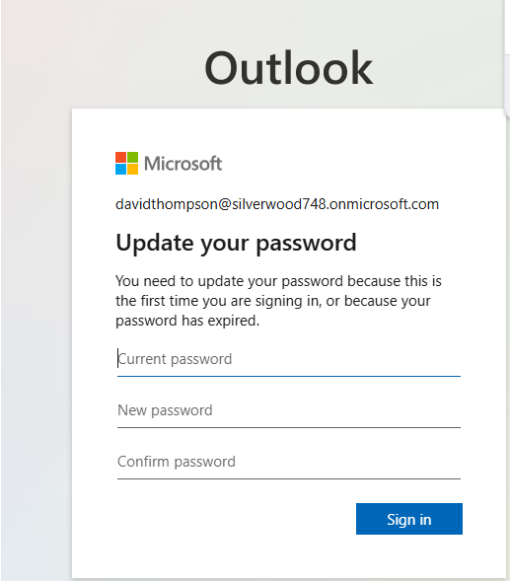
- You can see all created teams: *All Company*, *Instructors*, *IT*, *SilverWood*, *Staff*, and *Students*.
- For each team, the email address, sync status, privacy type, and creation time are listed.
- The **"Students"** team is selected, and the **member list** is shown on the right side, including:
  - *Alice Zhang, Brian Lee, Chloe Nguyen, Daniel Kim, Ella Chen, Fiona Li,* and others — demonstrating that students were correctly assigned to the group.

Outlook Login Process

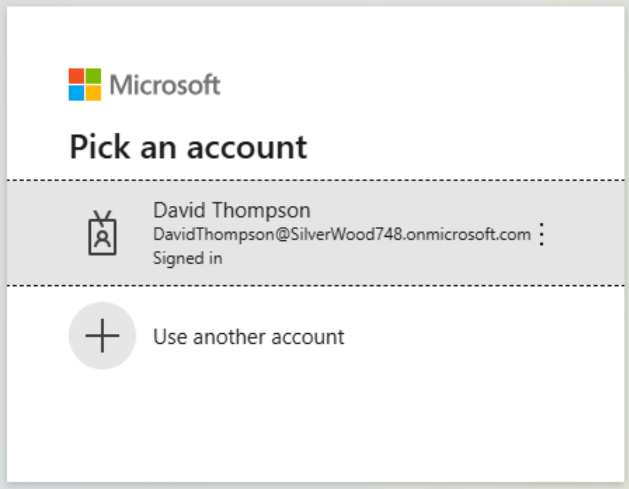
- These images document **David Thompson** logging into his Microsoft 365 email account through Outlook.
  1. **Initial Sign-In** using assigned email and temporary password.



  2. **Password Update Prompt** appears because it's the first login or password has expired.



3. **Account Selection Screen** confirms successful login and account recognition.



These screenshots finalize a comprehensive setup for Microsoft 365 group and user management in an educational organization:

- Four **Teams groups** were created: IT, Staff, Instructors, and Students.
- Each group has clearly assigned **owners**, **members**, and **private access settings**.
- The **Teams and Groups panel** confirms group creation and user assignments.
- Individual user access to Outlook is verified via the login and password update steps.

This workflow ensures a secure, organized, and role-based collaboration structure, tailored to a school or training institute's internal communication needs.

End Microsoft 365 Email & Team Management Setup

This entire process demonstrates how a Microsoft 365 environment was built for an academic or organizational structure, including:

- Account creation and role-based admin access
- Group (Teams) setup for Staff, IT, Instructors, and Students
- Assignment of owners and members
- Group privacy configuration (Private groups)
- Successful email communication:
    - Group messages
    - Individual reminders
    - Announcements
    - Replies and inbox tracking
- Role-based account sign-ins and first-time password reset flows

Microsoft Teams Configuration Summary

I configured Microsoft Teams to support group-based communication and collaboration within the SilverWood school domain. Here's what I did:

1. **Created and Managed Teams for Four Groups**
    - **Staff**
        - Owners: Lily Wong, Aiwei Tu
        - Members: James Chen, Michael Sun, Emma Lee

- **IT**
- Owners: Aiwei Tu, Milan Singh
- Members: Chigozirim Success Ibe, Limusi A

- **Instructors**
    - Owners: Aiwei Tu, Amanda White
    - Members: David Thompson, Cynthia Brown, Benjamin Clark

- **Students**
    - Owner: Amanda White
    - 15 members including Jason Wu, George Wang, OliviaLin, etc.

2. **Scheduled a Curriculum Meeting**
    - Meeting topic: **"Curriculum Meeting"**
    - Time: April 29, 2025, from 3:00 PM to 3:30 PM
    - Participants: **Students** and **Staff** teams
    - Meeting description: *"Talk about the curriculum of next term"*

3. **Verified Meeting Visibility and Delivery**
    - The meeting successfully appeared on the calendar.

Users (e.g., OliviaLin) received invitations and joined the appropriate Teams groups.

File Server

Successfully deployed and configured a **Windows Server 2022 virtual machine** named FileServer using **Hyper-V** on a host joined to the SilverWood.local domain. Here's what I accomplished:

1. **Virtual Machine Creation & OS Installation**
    - Created a Generation 2 VM named **FileServer** with 8192MB RAM.
    - Installed the Windows Server operating system from an ISO file.

Configuration

- Set the Administrator password.
- Configured a **static IP address**:
    - IP: 192.168.50.13
    - Subnet: 255.255.255.0
    - Gateway: 192.168.50.1
    - DNS: 192.168.50.11

Domain Integration

Successfully **joined the FileServer to the Active Directory domain** SilverWood.local.

Role Installation

Installed File Server Resource Manager via Server Manager to support advanced file services.

Although I did not perform an in-depth configuration of quotas or file screening, this step was intended to demonstrate my awareness of advanced file management tools available in Windows Server environments. It reflects my initiative to go beyond basic role setup and familiarize myself with resource control capabilities used in real-world server administration.

Shared Folder Setup & Permissions

- Created a shared folder named **Mylearning**.
- Assigned **share and NTFS permissions** for different groups:
    - IT_group: Full Control
    - Instructors_group: Modify
    - Staff_group: Read
    - Student_group: Read & Execute

User Credential Configuration

- Added credentials for domain user SILVERWOOD\OliviaLin to access the FileServer and it got denied.

Access Testing

- Logged in using the domain user account **Aiwei** from the IT group.
- Verified **read and write access to the shared folder**, confirming correct permissions.
- Created test files inside the shared folder from a networked machine, verifying access control: One file from a user with write access (test from aiwei)

GPO

Configured a Group Policy Object (GPO) to enforce a login message

Enabled policy: Interactive logon: Message text for users attempting to log on

Verified the policy was applied successfully by logging into the domain machine:

Displayed message: "Welcome to SilverWood! Unauthorized access is prohibited. Please contact IT if you have any issues."

6.0 SECURITY HARDENING

6.1 For Routers

- SSH ENABLED
- NO CDP
- MOTD BANNER
- ACLs for OSPF

For Switches

- **Unused Ports Shutdown**
- *Port-Security enabled Which limits devices that can connect*
- *And sticky MAC that binds a MAC to the port dynamically and adds it to config*
- *VLANs*
- **DHCP Snooping**
- **NO CDP**
- **SSH Enabled**
- **BPDU Guard**

6.2 SERVER HARDENING

- **Windows Automatic Updates**
- Windows defender real time protection , this Monitors your system in **real time** and blocks threats (viruses, malware, spyware, ransomware) **before** they cause damage.

- Windows Firewall Prevents **unauthorized inbound or outbound connections**, protecting systems from hackers or malicious software.
- Inbound Rules ; Shadow (TCP-In) , Usermode (TCP-in), Usermod (UDP-in)
- Enabling Windows Firewall on server
- Applying GPOs of sending notification to the IT admins on any unforeseen illegal action